This allows one to define the tensor product of any number of vector spaces, $V_1 \otimes \cdots \otimes V_n$. Note that this tensor product is associative, in the sense that $(V_1 \otimes V_2) \otimes V_3$ can be naturally identified with $V_1 \otimes (V_2 \otimes V_3)$.

In particular, people often consider tensor products of the form $V^{\bigotimes n} = V \otimes \cdots \otimes V$ (n times) for a given vector space V, and, more generally, $E := V^{\bigotimes n} \otimes (V^*)^{\bigotimes m}$. This space is called **the space of tensors of type** (m, n) on V. For instance, tensors of type $(0, 1)$ are vectors, tensors of type $(1, 0)$ — linear functionals (covectors), tensors of type $(1, 1)$ — linear operators, of type $(2, 0)$ — bilinear forms, tensors of type $(2, 1)$ — algebra structures, etc.

If V is finite dimensional with basis e_i, $i = 1, \ldots, N$, and e^i is the dual basis of V^*, then a basis of E is the set of vectors

$$e_{i_1} \otimes \cdots \otimes e_{i_n} \otimes e^{j_1} \otimes \cdots \otimes e^{j_m},$$

and a typical element of E is

$$\sum_{i_1,\ldots,i_n,j_1,\ldots,j_m=1}^{N} T^{i_1\ldots i_n}_{j_1\ldots j_m} e_{i_1} \otimes \cdots \otimes e_{i_n} \otimes e^{j_1} \otimes \cdots \otimes e^{j_m},$$

where T is a multidimensional table of numbers.

Physicists define a tensor as a collection of such multidimensional tables T_B attached to every basis B in V, which change according to a certain rule when the basis B is changed (derive this rule!). Here it is important to distinguish upper and lower indices, since lower indices of T correspond to V and upper ones to V^*. The physicists don't write the sum sign, but remember that one should sum over indices that repeat twice — once as an upper index and once as lower. This convention is called the *Einstein summation*, and it also stipulates that if an index appears once, then there is no summation over it, while no index is supposed to appear more than once as an upper index or more than once as a lower index.

One can also define the tensor product of linear maps. Namely, if $A : V \to V'$ and $B : W \to W'$ are linear maps, then one can define the linear map $A \otimes B : V \otimes W \to V' \otimes W'$ given by the formula $(A \otimes B)(v \otimes w) = Av \otimes Bw$ (check that this is well defined!). The